



Selected topics in Cryptography

Quiz 1A

February 26, 2026

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Problems

Read each of the following problems and solve them. Please include the detail of all your calculations

1. Write down the pseudocode of an algorithm that receives as input a prime number $p > 3$, and find parameters $a, b \in \mathbb{Z}_p$ of every non-singular elliptic curve over $\mathbb{Z}_p$. (Correct answer: 1 point)

2. Consider the following elliptic curves defined over $\mathbb{Z}_5$: $y^2 = x^3 + 2x + 2 \pmod{5}$ and $y^2 = x^3 + x + 2 \pmod{5}$. Choose the non-singular elliptic curve and do the following

- Compute all points for the non-singular elliptic curve E over $\mathbb{Z}_5$.
- How many points does E have?

(Correct answer: 4 points)

① Algor: Lmo 1
Input: un número primo $p > 3$
Output: $a, b \in \mathbb{Z}_p$ que forman EC no singulares sobre $\mathbb{Z}_p$
for $a \rightarrow 0$ to $p-1$:
 for $b \rightarrow 0$ to $p-1$:
 si $4a^2 + 27b^2 \neq 0$ entonces:
 $conj[] = conj[] + (a, b)$ → lista con las tuplas (a, b)
 que cumplen la condición
 end if
 end for
end for
return $conj[]$

② $E: y^2 = x^3 + x + 2 \pmod{5}$

$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

① $\mathbb{Q}R = \{0, 1, 4\}$

¿cómo sabes que esta es no singular?

$\mathbb{Q}R$	$S\mathbb{Q}$
0	0
1	± 1
4	± 2

x	$x^3 + x + 2 \pmod{5}$	y
0	2	-
1	4	± 2
2	2	-
3	2	-
4	0	0

$E(\mathbb{Z}_5) = \{(1, 2), (1, -2), (4, 0), \infty\}$

② $\#E = 4$